# MediFlow

## Hospital ERP + CRM

Phase 1 MVP — User, Demo & Operations Guide

This document explains what MediFlow is, which module to use when, how to run common workflows, role-based access (RBAC), demo logins, the in-app Help page, and deployment notes for Vercel demos.

Generated: 13/11/2026, 12:20:03 pm

Repository: https://github.com/rounakdutta121/mediflow

# 1. What is MediFlow?

MediFlow is a modern single-hospital operations platform that combines front-office CRM (patients, appointments, queue) with clinical EMR basics and OPD billing. It is built with Next.js, MongoDB Atlas, Auth.js, and role-based permissions so each staff role only sees and does what they should.

### Phase 1 includes

- Secure login, JWT sessions, RBAC, notifications, audit trail
- Patient registration with UHID, search, and profile
- Departments, doctors/staff directory, doctor schedules
- Appointment booking, check-in, token queue
- Clinical encounters, notes, vitals, diagnoses, signed prescriptions
- Service catalogue, invoices, payments/receipts, auto-invoice from priced Rx on sign
- Operational dashboard, MTD reports, hospital settings, Help guide
- Light/dark theme, responsive UI, mobile sidebar

### Later phases (navigation shells only)

- Admission & Beds
- Laboratory
- Pharmacy
- Insurance / TPA
- Nursing

# 2. Demo credentials (IDs & passwords)

Use these accounts after seeding the database (npm run seed). Passwords are case-sensitive.

| Role | Email / ID | Password | Best for |
|---|---|---|---|
| Hospital Admin | admin@mediflow.local | Admin@12345 | Full access, settings, staff |
| Doctor | dr.mehta@mediflow.local | Doctor@12345 | EMR, queue, own appointments |
| Doctor | dr.khan@mediflow.local | Doctor@12345 | Second clinician demo |
| Receptionist | reception@mediflow.local | Reception@12345 | Patients, booking, queue, billing |
| Accountant | accounts@mediflow.local | Accounts@12345 | Invoices, payments, reports |

Roles defined in code but not seeded by default: NURSE, AUDITOR. Admin can create them under Doctors & Staff.

### Demo hospital defaults (seed)

- Hospital name: MediFlow General Hospital

- Hospital code: MFH
- Timezone: Asia/Kolkata · Currency: INR
- Sample patient (if present): Rounak Dutta · UHID MFH-001006
- Sample services: OPD Consultation, Follow-up, ECG, CBC

**Security note: Change all demo passwords before any real/production use. Never commit .env.local.**

# 3. When to use which module

| Situation | Open this module | Who usually uses it |
|---|---|---|
| New walk-in / registration | Patients !' Register | Reception |
| Schedule a visit | Appointments / Patient !' Book | Reception |
| Patient arrived today | Queue !' Check-in | Reception / Nurse |
| Doctor starts consult | Queue / Clinical !' New | Doctor |
| Write Rx / vitals / notes | Clinical !' Encounter | Doctor |
| Collect money / OPD bill | Billing | Reception / Accounts |
| Month-to-date KPIs | Reports / Dashboard | Admin / Accounts |
| Investigate changes | Audit | Admin / Auditor |
| Hospital name, contact, TZ | Settings | Admin only |
| How to do workflows | Help (after Settings) | All roles |
| Add doctor / change role | Doctors & Staff | Admin |
| Add specialty units | Departments | Admin |

# 4. How to use — step by step

## 4.1 Sign in

- Open the app URL (local http://localhost:3000 or your Vercel URL).
- Enter email + password from the credentials table.
- Optional: use the theme toggle (sun/moon) for light or dark mode.
- Sidebar only shows modules your role can access.

## 4.2 Register a patient

- Go to Patients !' Register patient (needs patients:write).
- Fill first/last name and phone (required). Add allergies/conditions as comma lists.
- Save — system assigns a UHID (e.g. MFH-001007).
- From profile: Book appointment, Start encounter, Create invoice (only if your role allows).

## 4.3 Book an appointment

- Appointments !' Book appointment, or Patient !' Book appointment.
- Select patient, doctor, department, date, time, type (OPD / follow-up / walk-in / teleconsult).
- Save. Appointment appears on the appointments board for that date.

## 4.4 Check-in & queue

- Open Queue (queue:manage).
- Check in a scheduled patient for today — token is generated.
- Doctors with clinical:write see Start consult; reception typically only manages queue status.

### 4.5 Clinical encounter (EMR)

- Clinical !’ New encounter, or Start encounter from queue/patient.
- Document chief complaint, history, examination, assessment, plan, vitals, diagnosis.
- Add prescriptions with optional service/price; signing a priced Rx can auto-create an invoice.
- Edit notes/vitals while open; delete only with clinical:write.

### 4.6 Billing & payments

- Billing !’ Create invoice: pick patient, add catalogue services, discount, save.
- Open invoice !’ Collect payment (cash/card/UPI etc. as available in form).
- Status moves draft/issued !’ partially paid !’ paid. Balance shows on dashboard.

### 4.7 Staff, departments, settings

- Doctors & Staff: create users with role + password; edit/deactivate (staff:write = Admin).
- Departments: create/edit/delete active departments (departments:write = Admin).
- Settings: edit hospital name, code, address, phone, email, timezone, currency (settings:manage = Admin).

### 4.8 Help page (in app)

- Open Help from the sidebar (listed after Settings).
- Use it as an on-screen checklist for register !’ book !’ check-in !’ consult !’ bill.
- All signed-in roles can open Help; actions still respect RBAC on each module.

### 4.9 Notifications

- Bell icon in the top bar lists targeted alerts (patient, appointment, clinical, billing, system).
- Hospital Admin receives all notifications; others receive role-targeted ones.
- Mark one read, mark all read, or delete from the panel.

## 5. RBAC — roles & permissions

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Every page and API is permission-gated. Missing permission !’ UI hides the action; direct URL redirects to dashboard?error=forbidden; API returns 403.

### 5.1 Permission catalogue

- dashboard:view — home KPIs
- patients:read / patients:write — view vs register/update/delete
- appointments:read / appointments:write — list vs book/update/cancel
- queue:manage — check-in and queue board
- staff:read / staff:write — directory vs create/edit staff
- departments:read / departments:write — view vs manage departments
- clinical:read / clinical:write — view EMR vs create/edit/sign/delete
- billing:read / billing:write — view invoices vs create/pay/delete
- reports:view — analytics
- audit:view — audit log
- settings:manage — hospital configuration

### 5.2 Role matrix

| Capability | Admin | Reception | Doctor | Nurse | Accounts | Auditor |
|---|---|---|---|---|---|---|
| Dashboard | Yes | Yes | Yes | Yes | Yes | Yes |
| Patients write | Yes | Yes | No | No | No | No |
| Appointments write | Yes | Yes | No | No | No | No |
| Queue | Yes | Yes | Yes | Yes | No | No |
| Clinical write | Yes | No | Yes | No | No | No |

| Capability | Admin | Reception | Doctor | Nurse | Accounts | Auditor |
|---|---|---|---|---|---|---|
| Billing write | Yes | Yes | No | No | Yes | No |
| Reports | Yes | No | No | No | Yes | Yes |
| Audit | Yes | No | No | No | No | Yes |
| Staff write | Yes | No | No | No | No | No |
| Settings | Yes | No | No | No | No | No |
| Help page | Yes | Yes | Yes | Yes | Yes | Yes |

# 6. Recommended demo script (5–10 minutes)

- 1. Login as Admin — show dashboard KPIs, toggle dark mode, open notifications and Help.
- 2. Switch to Reception — register a patient, book appointment with Dr Mehta, check-in on Queue.
- 3. Switch to Doctor — open Queue/Clinical, start encounter, add vitals + Rx with price, sign.
- 4. Switch to Accounts — open Billing, confirm invoice / collect payment, open Reports.
- 5. Back to Admin — show Audit trail and editable Hospital settings.

# 7. Local & Vercel setup

## Local

- npm install
- Copy .env.example !’ .env.local and set MONGODB_URI + AUTH_SECRET + AUTH_URL
- npm run seed
- npm run dev !’ http://localhost:3000

## Vercel demo

- Import GitHub repo rounakdutta121/mediflow
- Set env: MONGODB_URI, AUTH_SECRET, AUTH_URL (your https://….vercel.app)
- Deploy, then run seed once against the same Atlas URI
- Atlas network access must allow Vercel (0.0.0.0/0 is typical for demos)

# 8. Support & scope notes

- Phase 1 is single-hospital OPD-focused CRM/ERP foundation.
- Admission, lab, pharmacy, insurance, and nursing are placeholders for later phases.
- Soft-delete is used for many entities (e.g. patients marked inactive via Delete in UI).
- For demos, keep seeded credentials handy; rotate them for any shared public deployment.

**End of MediFlow Phase 1 User & Demo Guide**

MediFlow — Clinical ops, flowing with clarity.